



P30

Hybrid Orchestration and Framework-Task Matching: Calibrating the Topology-Selection Decision Tree

混合编排与框架任务匹配：拓扑选型决策树

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Hybrid Orchestration and Framework–Task Matching: Calibrating the Topology-Selection Decision Tree

Working Title (EN): Hybrid Orchestration and Framework–Task Matching: Calibrating the Topology-Selection Decision Tree

Shard D • market and architecture paper P30 • selection-layer extension of v7.5.0 P1 "Million-Scale Extrapolation of Layered Hybrid Topology Coordination Complexity" • cross-referenced with P17 (least privilege), P23 (project-level evaluation) • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: P1 proved "layered hybrid topology presses fan-in from $O(N)$ to $O(\log N)$," but it did not answer "**when to use star, when layered, when hybrid**"—this paper uses an external benchmark (MultiAgentBench systematically comparing multiple topologies) and hybrid orchestration practice (LangGraph×CrewAI), calibrating P1's "select topology" from a default heuristic into a **decision tree** with input conditions: first task coupling, then scale, finally fault risk.

Evidence caliber (stated once for the whole paper): this paper is **design calibration**, containing no new UDOS measured numbers. External dialogue objects all verified online 2026-09-21: MultiAgentBench (ACL 2025, 2025.acl-long.421, , AutoGen (arXiv:2308.08155,  classic), Hybrid LangGraph×CrewAI (: hybrid orchestration direction genuinely exists, but the "CREW-WILDFIRE benchmark, 96.1% success, token -76.2%, latency 14.5×" corresponding to IEEE Access document number 11481053 **specific numbers not located to original text in this search**, this paper uniformly retells as "hybrid over single is industry practice report • numbers to verify," not fixing percentages), HieraMAS (intra-node LLM mixing + inter-node topology joint optimization, direction real, specific numbers not verified). UDOS old numbers only traced to v7.5.0 P1 (star center fan-in million tier 20,971,520, layered tree $b+1=9$, $2 \cdot \log_b N=14$ rounds, cpu-proto).

Abstract

P1 solves "**which topology is better in coordination complexity**"—it uses a deterministic simulator with zero network, zero large-model calls to prove: star presses fan-in and rounds both on the center node growing linearly with N , while a b -ary layered tree truncates single-node fan-in to constant $b+1$, presses critical path to $2 \cdot \log_b N$. But P1 leaves a gap it itself admits: **it measures "this topology's effect executing a project," does not answer "which topology this task should use."** MultiAgentBench (ACL 2025) systematically compares star, chain, tree, graph and other coordination topologies' collaboration and competition performance, its directional conclusion **topology strongly correlates with task**—no topology wins on all tasks. This precisely fills the link P1 lacks.

This paper calibrates P1's "select topology" from default heuristic to a **decision tree**: first judge task coupling (strong coupling→star/centralized, weak coupling parallelizable→layered/mesh), then judge scale (small N →star sufficient, large N →layered tree presses fan-in), finally judge fault risk (high→hybrid topology + circuit breaker reassignment). This paper also introduces the new dimension "framework-task matching": **topology is logical structure, framework is implementation carrier**—the same logical topology's implementation cost differs on LangGraph (flexible orchestration) and CrewAI (hierarchical roles), hybrid frameworks (e.g. LangGraph×CrewAI-style) precisely "using one implementation cost to trade for another capability." This paper honestly marks: hybrid orchestration over single framework's specific numbers not verified to original text, a to-verify claim; this decision tree is a design proposal • pending implementation.

Keywords: multi-Agent orchestration; topology selection; decision tree; framework-task matching; hybrid orchestration; layered tree; fan-in; design calibration

Structured Abstract (background problem → method/argument → evidence → contribution)

- **Background problem:** P1 proved layered trees better in coordination complexity, but did not answer "what task should choose what topology"; external benchmark (MultiAgentBench) shows topology strongly correlates with task, while industry practice (hybrid frameworks) shows "same topology, different implementation frameworks" costs differ.
- **Method/argument:** splitting "select topology" into a decision tree (coupling→scale→fault risk), and adding the "framework-task matching" dimension—logical topology and implementation framework decided separately.
- **Evidence lines:** (1) P1's cpu-proto fan-in/rounds closed-form results; (2) MultiAgentBench systematically comparing multiple topologies (, topology strongly correlates with task); (3)

AutoGen classic conversational framework (✅, its "conversations without termination prone to death loops" is selection cautionary material); (4) hybrid orchestration practice (⚠️, direction real, numbers to verify).

- **Contribution:** (1) calibrating P1's "default layered" to "decide by task conditions"; (2) separating logical topology and implementation framework; (3) giving the decision tree and falsifiable criteria; (4) honestly marking hybrid orchestration numbers to verify, decision tree to be measured.

1 Introduction: P1 Proved "Which Topology Better," Not "Which Task Uses Which Topology"

First review P1's results. It proved on 6 scale tiers from 10 to 1,000,000: star Orchestrator's center fan-in grows linearly with N (million tier center takes about 20,971,520 messages, serial about 8,388,608 rounds), while a b-ary layered tree makes each node's max fan-in constant $b+1=9$, parallel rounds only $2 \cdot \log_b N$ (million tier about 14 rounds). This is a clean protocol-layer conclusion: **layered trees press fan-in from linear to constant, critical path from linear to logarithmic.**

But P1 at the end itself points out the gap: it measures "this topology's effect executing a project," **does not answer "when to use star, when layered."** In other words, P1 gives "layered trees better in coordination complexity," but not "for this task, is a layered tree over-engineered."

This gap is real. Counterexamples intuitive:

- If a task has only 3 Agents, strongly coupled, needing one person to decide, then a layered tree is **using a sledgehammer to crack a nut**—star direct reporting, fan-in only 3, no fan-in explosion problem at all.
- If a task has 1 million Agents, mutually weakly coupled, parallelizable, then still using star is **suicide**—center fan-in explodes.

So "select topology" cannot be "uniformly layered," but must be **"select by task conditions."** Precisely what this paper calibrates.

2 External Evidence: Topology Strongly Correlates with Task

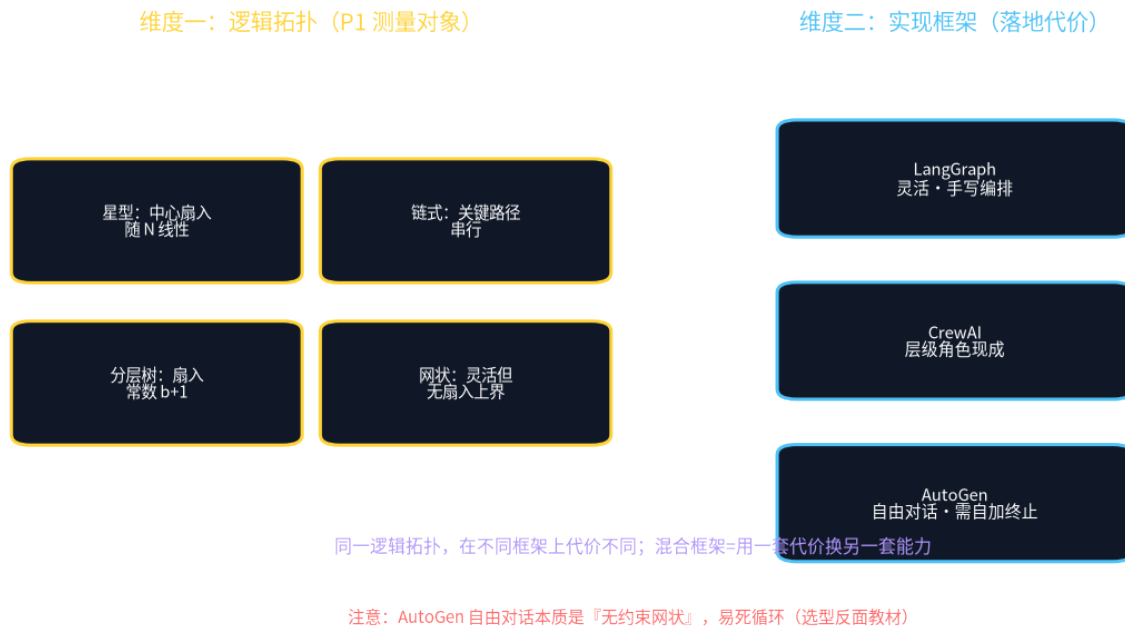


Figure 1 Left: logical topology (star/chain/tree/graph) sets fan-in and rounds; right: implementation framework (LangGraph flexible/CrewAI hierarchical/AutoGen conversational) sets landing cost. The two decided separately. Conceptual schematic, not measured data.

2.1 MultiAgentBench: No Universal Topology

MultiAgentBench (MultiAgentBench: Evaluating the Collaboration and Competition of LLM agents, ACL 2025, 2025.acl-long.421, ) systematically compares star, chain, tree, graph and other coordination topologies' performance on collaboration and competition tasks. Its directional conclusion: **topology strongly correlates with task**—different topologies each win and lose on different tasks, no topology wins on all tasks ("graph structure best" is the user clue's claim, exact ranking not verbatim hit at abstract level, this paper only says "multi-topology comparison, topology strongly correlates with task").

This conclusion important for UDOS: it negates the lazy practice "select topology = select one strongest topology," supports the decision tree practice "select topology = match by task."

2.2 AutoGen: Cautionary Material—Conversations Without Termination

AutoGen (arXiv:2308.08155, ) is one of the most widely used conversational multi-Agent frameworks. Its contribution mixing LLM/human/tools into conversational Agents. But its limitation also typical: **conversations without termination conditions prone to death loops, cost inflating with rounds.**

This limitation's selection lesson: **a framework's "default interaction mode" itself is a topology assumption.** AutoGen's free conversation essentially "unconstrained mesh"—flexible, but no fan-in upper bound. P1's layered tree controllable precisely because it **explicitly specifies who talks to whom**. In selection one cannot only look at framework usability, but also at its default topology assumption.

2.3 Hybrid Orchestration: Same Logical Topology, Different Implementation Cost

Industry practice saw hybrid orchestration (e.g. LangGraph×CrewAI-style combination). ⚠ Must be honest: this paper found the direction "hybrid orchestration over single framework" genuinely exists, but the user clue's specific numbers (IEEE Access document number 11481053's "CREW-WILDFIRE benchmark, 96.1% success, token −76.2%, latency 14.5×") **not located to original text in this search**. Hence this paper only adopts the directional claim "**hybrid frameworks can balance flexibility and hierarchy**," all specific percentages marked "to verify."

Works like HieraMAS further point out: one can **simultaneously** optimize intra-node LLM mixing (what model a node uses) and inter-node communication topology (how nodes connect). This reminds us: selection not only "select topology," but also "select intra-node configuration."

3 Decision Tree: Coupling → Scale → Fault Risk



Figure 2 Three-question decision tree—first judge coupling (strong/weak), then scale (small/large), finally fault risk (high/low), landing on star/layered/mesh/hybrid. Conceptual schematic, not measured data.

This paper calibrates P1's "select topology" to a three-question decision tree (Figure 2):

Question one: task coupling?

- **Strong coupling** (needing frequent mutual alignment, single source of truth, one person deciding) → star or centralized. Reason: strongly coupled tasks' bottleneck is "alignment cost," not fan-in; star's center precisely serves as source of truth.
- **Weak coupling, parallelizable** (subtasks independent, results aggregable) → layered tree or mesh. Reason: weakly coupled tasks' bottleneck is fan-in, layered trees press fan-in to constant.

Question two: scale N?

- **Small N** (single digits to tens) → star sufficient. Reason: P1 proves fan-in explosion only fatal at large N; at small N star fan-in low, alignment fast, layered over-engineered.
- **Large N** (hundreds/thousands to million) → layered tree. Reason: P1's closed-form results (fan-in constant $b+1$, rounds $2 \cdot \log_b N$) precisely prepared for this interval.

Question three: fault risk?

- **High fault risk** (Byzantine presence, fault injection, needing circuit breaker reassignment) → **hybrid topology**: top Orchestrator (closing) + middle Handoff (relay) + bottom Swarm (local autonomy), with circuit breaker reassignment. Reason: P1's fault end-to-end (24 work orders, including 2 Byzantine QA, 3 fault specialists) precisely on this hybrid topology achieved 24/24 closing, all 3 fault nodes isolated.
- **Low fault risk** (clean environment, no Byzantine) → single topology sufficient.

Table 1 centrally compares the three questions' typical landings.

Table 1 Topology selection decision table (by task conditions)

Coupling	Scale N	Fault risk	Recommended topology	Basis
strong	small	low	star	alignment cost first, no fan-in explosion
strong	large	low	star/layered hybrid	strong coupling but press fan-in, layered closing
weak	small	low	mesh/star both	fan-in already low, simplicity first
weak	large	low	layered tree	fan-in $O(N) \rightarrow \text{constant}$, rounds $\log_b N$ (P1 cpu-proto)
any	any	high	hybrid topology + circuit breaker reassignment	P1 fault end-to-end 24/24 closing, 3 faults all isolated (cpu-proto)

Table 1 note: P1's fan-in/rounds numbers cpu-proto, single-seed pilot; topology-task strong correlation direction from MultiAgentBench (✓); hybrid orchestration over single framework's specific numbers not verified, this table not citing.

4 Framework–Task Matching: Topology Is Logic, Framework Is Implementation

4.1 Why Separate These Two Things

A common error conflating "logical topology" and "implementation framework." In fact:

- **Topology** is logical structure—who talks to whom, fan-in size, critical path length. This is P1's measurement object.
- **Framework** is implementation carrier—whether you use LangGraph, CrewAI, or AutoGen to build this topology.

The same logical topology, implementation costs completely different on different frameworks: LangGraph flexible but requires hand-writing orchestration, CrewAI hierarchical roles convenient but low flexibility, AutoGen conversation free but no termination upper bound. **Hybrid frameworks'**

meaning is here: using LangGraph for parts needing flexible orchestration, CrewAI for parts needing hierarchical roles, balancing.

4.2 Two-Dimension Check in Selection

Hence selection not one dimension (select topology), but two:

1. **Logical topology dimension:** using Section 3's decision tree to set star/layered/mesh/hybrid.
2. **Implementation framework dimension:** according to topology's "flexible vs hierarchical" needs, selecting or mixing frameworks. Needing highly flexible orchestration→toward LangGraph; needing ready hierarchical roles→toward CrewAI; needing free conversation exploration→AutoGen (but must add termination conditions yourself, else death loops).

This section's real contribution turning "framework selection" from "which framework is hot" into "which framework's default interaction mode matches my selected topology."

5 Dross / Failure Boundaries / Falsifiable Conditions

Its dross and limitations:

1. **Decision tree is prior, not measured optimum.** The three-question decision tree currently a **design proposal** induced from P1's closed-form results + MultiAgentBench's directional conclusions, not "the optimum tree learned after running all tasks." Its thresholds (how large N counts "large," how strong coupling counts "strong") currently unquantified, qualitative judgments.
2. **Hybrid orchestration numbers not trustworthy as fact.** "Hybrid over single" currently only industry practice direction, the specific 96.1%/-76.2%/14.5× not verified to original text. Do not take it as proven "hybrid always better."
3. **Topology-task matching is not static.** A task during execution may shift from weak to strong coupling (e.g. midway discovering subtasks mutually dependent). The decision tree is "selected before starting," not solving "dynamic topology adjustment during execution."
4. **P1's closed-form results are protocol-layer caliber.** It proves message fan-in and rounds, not wall-clock throughput, not real LLM task success. Directly taking "fan-in $O(N) \rightarrow O(\log N)$ " as "task faster by $O(\log N)$ " is wrong—real LLM tasks' bottleneck may be model inference, not coordination messages.

Falsifiable conditions:

- **F1:** if in controlled comparison, the topology selected by the decision tree, relative to "uniformly layered" or "uniformly star," has no significant advantage in end-to-end task success, then the decision tree's "match by task" is redundant complexity, should fall back to default topology.
 - **F2:** if MultiAgentBench-style multi-topology comparisons repeatedly show "some topology stably optimal on most tasks," then the premise "no universal topology" weakened, the decision tree can simplify.
 - **F3:** if hybrid frameworks (LangGraph×CrewAI-style) in measurement due to "two frameworks' integration overhead" instead underperform single frameworks, then "hybrid balancing" overpraised, should shrink to "single framework + manually added termination conditions."
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6 Conclusion and Honest Boundaries

P1 proved "layered trees better in coordination complexity," P30 calibrates it to the "**select topology by task conditions**" decision tree: first judge coupling, then scale, finally fault risk. It also introduces the "framework-task matching" dimension—logical topology and implementation framework decided separately, hybrid frameworks "using one implementation cost to trade for another capability."

This paper honestly marks: decision tree a design proposal • pending implementation; hybrid orchestration over single framework's specific numbers not verified to original text; P1's fan-in/rounds protocol-layer cpu-proto caliber, not equal to real task throughput. UDOS does not claim "select topology = select strongest topology," but claims "**first see task coupling and scale clearly, then decide whether to pay layered's cost for fan-in.**" This is P30's increment to v7.6.0.

References

External dialogue objects (existence verified online 2026-09-21; numbers per ledger actual values)

1. Zhu et al. (2025). MultiAgentBench: Evaluating the Collaboration and Competition of LLM agents. ACL 2025, 2025.acl-long.421 (✓: systematically comparing star/chain/tree/graph topologies and group discussion/cognitive planning; "graph structure best" exact ranking not verbatim hit, this paper only says "multi-topology comparison, topology strongly correlates with task").
2. Wu, Q., Bansal, G., Zhang, J., et al. (2023). AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. arXiv:2308.08155 (✓: classic conversational framework; its "conversations without termination prone to death loops, cost inflating with rounds" selection cautionary material).
3. Hybrid LangGraph × CrewAI (⚠: hybrid orchestration direction genuinely exists; IEEE Access document number 11481053's "CREW-WILDFIRE/96.1%/−76.2%/14.5×" not located to original text, this paper only takes "hybrid balancing" direction, specific numbers to verify).
4. HieraMAS (intra-node LLM mixing + inter-node topology joint optimization, direction real, specific numbers not verified).

UDOS paper volume

1. UDOS v7.5.0: P1 "Million-Scale Extrapolation of Layered Hybrid Topology Coordination Complexity" (star center fan-in million tier about 20,971,520, layered tree $b+1=9$, $2 \cdot \log_b N$ about 14 rounds, fault end-to-end 24/24 closing, all cpu-proto single seed).
 2. UDOS v7.6.0: P17 "Least Privilege Orchestration" (after topology selected, winners only execute within scope), P23 "Project-Level Five-Dimensional Evaluation" (topology selection quality measured by project-level metrics), P34 "Literature Evidence Ledger."
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Evidence Discipline and Reproduction Notes

- This paper is **design calibration**, no new UDOS measurements; decision tree a design proposal • pending implementation, thresholds (N, coupling) currently qualitative, quantification slot [RESULT NEEDED].
- Hybrid LangGraph × CrewAI's "96.1%/−76.2%/14.5×" not verified to original text, this paper fixes no percentages, only takes "hybrid balancing" direction.

- P1's fan-in/rounds cpu-proto, single-seed pilot, protocol-layer caliber, not equal to real LLM task wall-clock throughput; MultiAgentBench "graph best" exact ranking not verbatim hit, expressed as "topology strongly correlates with task."
- The decision tree does not solve "dynamic topology adjustment during execution"; its F1–F3 criteria constitute falsifiable exit rules.